

Linear Algebra — Exercises

Complex numbers

Exercise 1. Calculate: a) $(2+3i)(4-2i)$ b) $(1+2i)(2i-1)$ c) $(3+i)(3-i)$ d) $(a+bi)(a-bi)$

Exercise 2. Calculate: a) $\frac{2-3i}{3+4i}$ b) $\frac{i}{1+i}$ c) $\frac{3-4i}{i}$

Exercise 3. Calculate: a) $\frac{1+4i}{4-i}$ b) $\frac{(3+4i)^4}{(3-4i)^3}$

Exercise 4. Show that for every two complex numbers z and w :

a) $\overline{z+w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z} \bar{w}$

b) $|zw| = |z| |w|$

c) $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$

Exercise 5 (*). Show that multiplication of complex numbers is associative.

Exercise 6 (*). Show that multiplication of complex numbers is distributive with respect to addition.

Exercise 7. Describe geometrically the sets of points $z \in \mathbb{C}$ satisfying the following conditions:

a) $|z - i + 3| = 5$

e) $\Re((z-i)^2) = 0$

b) $|z - i + 3| > 5$

f) $\Re(z^2) = 2$ and $(\Im(z+i))^2 = 1$

c) $|z - 1 + 3i| \leq 5$

g) $\left| \frac{z+3}{z-2i} \right| \geq 1$

d) $2 \leq |z+i| < 4$

h) $\Re(z+1) < 0$ and $|i-z| \leq 3$

Exercise 8. State which of the following are groups: a) $(\{z \in \mathbb{C} \mid |z| = 1\}, +)$

b) $(\{z \in \mathbb{C} \mid |z| = 1\}, \cdot)$

Exercise 9. Find polar forms of the following complex numbers:

a) $1+i$

g) $1-i\sqrt{3}$

b) -3 (negative 3)

h) $-z$, if $z = \cos \alpha + i \sin \alpha$

c) $-i$

i) $\sqrt{3} - i$

d) $2+2i$

j) $\bar{z}$, if $z = \cos \alpha + i \sin \alpha$

e) $3+3\sqrt{3}i$

k) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

f) $z = \sin \alpha + i \cos \alpha$

Exercise 10. Calculate:

a) $\sqrt[3]{(1+i)^6}$

d) $\left(\frac{2+2i}{\sqrt{3}-i} \right)^8$

b) $\left(\frac{1-i}{1+i} \right)^{222}$

e) $\sqrt[6]{1}$

c) $\left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)^{124}$

f) $\sqrt[4]{-1}$

g) $\sqrt[4]{(3-5i)^4}$

h) $\sqrt[6]{-1}$

Exercise 11. Describe geometrically the sets of all the following roots: a) $\sqrt[6]{1}$ b) $\sqrt[4]{-1}$ c) $\sqrt[8]{1}$
d) $\sqrt[6]{-1}$

Exercise 12. Show that for every n all complex roots of 1 of order n form a group under multiplication. Write down the table of the group of all complex roots of 1 of order 4.

Exercise 13. Use De Moivre's formula to:

- (a) express $\sin 5\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$,
- (b) express $\cos 4\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$.

Exercise 14 (*). Prove that

$$\frac{\cos \alpha + i \sin \alpha}{\cos \beta + i \sin \beta} = \cos(\alpha - \beta) + i \sin(\alpha - \beta).$$

Exercise 15. Find the polar form of: a) $2 + \sqrt{2} + i\sqrt{2}$ b) $2 + i + i\sqrt{3}$

Exercise 16. Calculate: a) $\cos \frac{\pi}{12}$ b) $\cos 105^\circ$ c) $\cos 75^\circ$ d) $\cos(-15^\circ)$

Exercise 17 (*). Calculate $(1 + \cos \alpha + i \sin \alpha)^{10}$ for $\alpha \in [0, \pi)$.

Exercise 18. Let $f(x) \in \mathbb{R}[x]$. Show that z is a root of f if and only if $\bar{z}$ is a root of f .

Exercise 19. Factor each of the following polynomials into a product of factors of degree at most 2:
a) $x^6 + 1$ b) $x^4 + 1$ c) $x^6 + x^3 + 1$ d) $x^5 + 1$

Exercise 20. Prove that every polynomial of an odd degree, with all real coefficients, has at least one real root.

Exercise 21. Knowing that $1 - 2i$ is a root of the polynomial $z^4 + 2z^3 + 2z^2 + 10z + 25$, find all remaining roots.

Exercise 22. Knowing that $2 + i$ is a root of the polynomial $z^4 - 2z^3 - 6z^2 + 22z - 15$, find all remaining roots.

Exercise 23. Solve the equations: a) $x^2 + x + 1 = 0$ b) $2x^2 + ix + 3 = 0$ c) $x^2 + 2x - i\sqrt{3} = 0$